

SSBMF Status Update: Merged TCGA + CPTAC Fit (April 30, 2026)

Andrew Walther

1. Original Problem and Proposed Solution

On the merged **TCGA_PAAD RNA-seq + CPTAC proteomics** training cohort ($n = 273$), SSBMF repeatedly returned $\hat{\beta}_k = 0$ for all factors. The model reconstructed the expression matrix but learned no survival signal, so risk prediction reduced to a constant hazard.

The first proposed fix was a **pre-processing reorder**. The original merged workflow selected the top 2,000 variable genes within each cohort *before* intersecting gene universes, leaving only about **838 shared genes**. The v2 pipeline instead intersected raw gene universes first, applied $\log_{2(x+1)}$ to RNA-seq only, quantile-normalized the merged samples, selected the top 2,000 genes by merged variance, and then rank-transformed each sample. This increased the shared analyzed feature set from **838 to 2,000 genes**.

Additional methods were then tested to determine whether the failure was caused by weak signal, poor initialization, or a structural issue in the CAVI updates.

Method tested	Quantitative result	Interpretation
λ sweep	12 combinations tested; no informative non-zero β recovered	Not a useful rescue path
High λ	At $\lambda \geq 5$, entire loading matrix collapsed	Over-amplified survival term
EBMF diagnostic	5/20 factors survival-associated; top C-index 0.629	Signal exists in merged data
EBMF warm-start, β only	6/20 active factors	β update is functional
EBMF warm-start, full CAVI	0/20 active after 23 iterations	Joint updates erase signal

Interpretation. Taken together, these tests showed that the merged failure was **not** caused by missing survival signal and **not** fixed by simple λ or prior tuning. Cluster A therefore focused on four in-model updates: (1) inner-loop reorder ($\beta \rightarrow L \rightarrow F$), (2) β -only burn-in ($N_{\text{burnin}} = 10$ in the smoke fit), (3) an α warmup/ramp schedule, and (4) survival/genomics rebalancing in the L update (`normalize_AB`).

2. Current Results and Issues Resolved

- **Pre-processing bug fixed:** the merged v2 pipeline increased the shared feature set from **838 to 2,000 genes** and quantile-normalized the merged cohort before variance filtering.
- **Training-side $\beta = 0$ collapse fixed:** Cluster A modifications produced a stable smoke fit with **2/20 active factors**, $\max |\hat{\beta}| = 0.0588$, **non-collapsed loadings** ($\max |EL| = 1.83 \times 10^3$), and **monotone ELBO** over 60 iterations.

Model behavior. This is an important change in model behavior. Before Cluster A, the merged fit converged to a degenerate solution with no prognostic factors at all. After Cluster A, the model no longer collapses immediately to the all-zero survival solution, and the ELBO remains well-behaved throughout the smoke fit.

Benchmark context. The existing **single-cohort** models already show that SSBMF can generalize externally when trained on one platform at a time. With the `point_normal` prior, **TCGA-only** training ($n = 144$) reached median external **C-index 0.602**, and **CPTAC-only** training ($n = 129$) reached median external **C-index 0.628**. The strongest single-cohort external result was on **PACA_AU_seq**, with **0.632** for TCGA-only training and **0.667** for CPTAC-only training. These results are not the joint-platform solution we want, but they provide a concrete benchmark for useful cross-cohort performance.

Why Cluster A worked:

- The **inner-loop reorder** lets the survival coefficient update fire before the loading update, so the survival term is no longer forced to start from a stale zero state.
- **Burn-in** gives β a chance to move away from zero before the joint CAVI loop can wash out survival information.
- `normalize_AB` addresses the key scale imbalance in `update_L_k()`: on the merged cohort, the survival precision term was roughly **0 to 2×10^{-4}** of the genomics term at iteration 1, so survival had almost no leverage on the loading update without rescaling.

Signal check. Unsupervised EBMF on the same merged matrix found **5/20 survival-associated factors**; the strongest achieved **C-index 0.629** with $p = 3 \times 10^{-6}$. The data therefore contain recoverable signal, and the failure is in the current SSBMF optimization path rather than in data quality alone.

3. Persisting Issues and Next Steps

The main unresolved issue is **generalization**. Although Cluster A recovered non-zero β during training, external validation was weaker than the **pre-Cluster A baseline fit** on most cohorts. Here, “baseline” refers to the merged v2 model **before** the Cluster A updates, i.e. the earlier fit without the new burn-in and `normalize_AB` changes:

- Improved on **1/5** external cohorts: Moffitt (**0.541 vs. 0.529; +0.012**).
- Regressed on **4/5** cohorts: Dijk (**0.606 vs. 0.626; -0.019**), PACA-AU array (**0.576 vs. 0.636; -0.060**), PACA-AU seq (**0.629 vs. 0.653; -0.024**), Puleo (**0.550 vs. 0.626; -0.076**).

Interpretation. This suggests the fix recovered factors aligned to the training Cox objective, but not factors that transport robustly across cohorts or platforms.

Practical implication. Cluster A solved the **training-side optimization failure** but did not solve the **cross-cohort generalization problem**. The current model can now produce non-zero survival coefficients on the merged training set, but those coefficients do not yet improve external prognostic performance consistently. The practical goal for merged TCGA+CPTAC training is therefore to retain the broader training set while moving external performance toward the single-cohort benchmark range of roughly **0.60 to 0.63 median C-index**.

Recommended next steps:

- Move the next development cycle to a new **cox-on-YF-reformulation** branch.
- Implement **Cluster B: Cox-on-YF**, replacing the survival predictor $\eta = L\beta$ with $\eta = (YF)\beta$, so training and prediction use the same observed-data projection.
- Derive and test the updated $q(F)$, $q(L)$, and $q(\beta)$ updates before additional benchmark runs.

Current blocker: Cluster B requires fresh derivations and new tests, so the next phase is a model reformulation rather than a tuning exercise.